Digital Lending Portfolio Optimization

A recommendation-led credit policy for sustainable, risk-adjusted growth

Recommendation

Restrict, Reprise , Remove - cutting 90+ default without losing profitable growth.

Contain the loss-makers by exiting out of the unfixable and catching the rest as they deteriorate.

1. Contain the tail of SME

Decline new SME C/D/E applications with weak cash-flow consistency.

2. Tighten personal Subprime

Decline Personal D/E applicants below the 0.6 cash-flow cut-off.

3. Cease BNPL

Digital ads + DSA - Stop acquisition on the two loss-making channels; existing book runs off.

4. Grow SME Prime

The offensive lever; replaces ~80% of shed volume.

5. Early warning Monitoring

The lifecycle gate: catches 87% of operating-book defaults ~104 days ahead

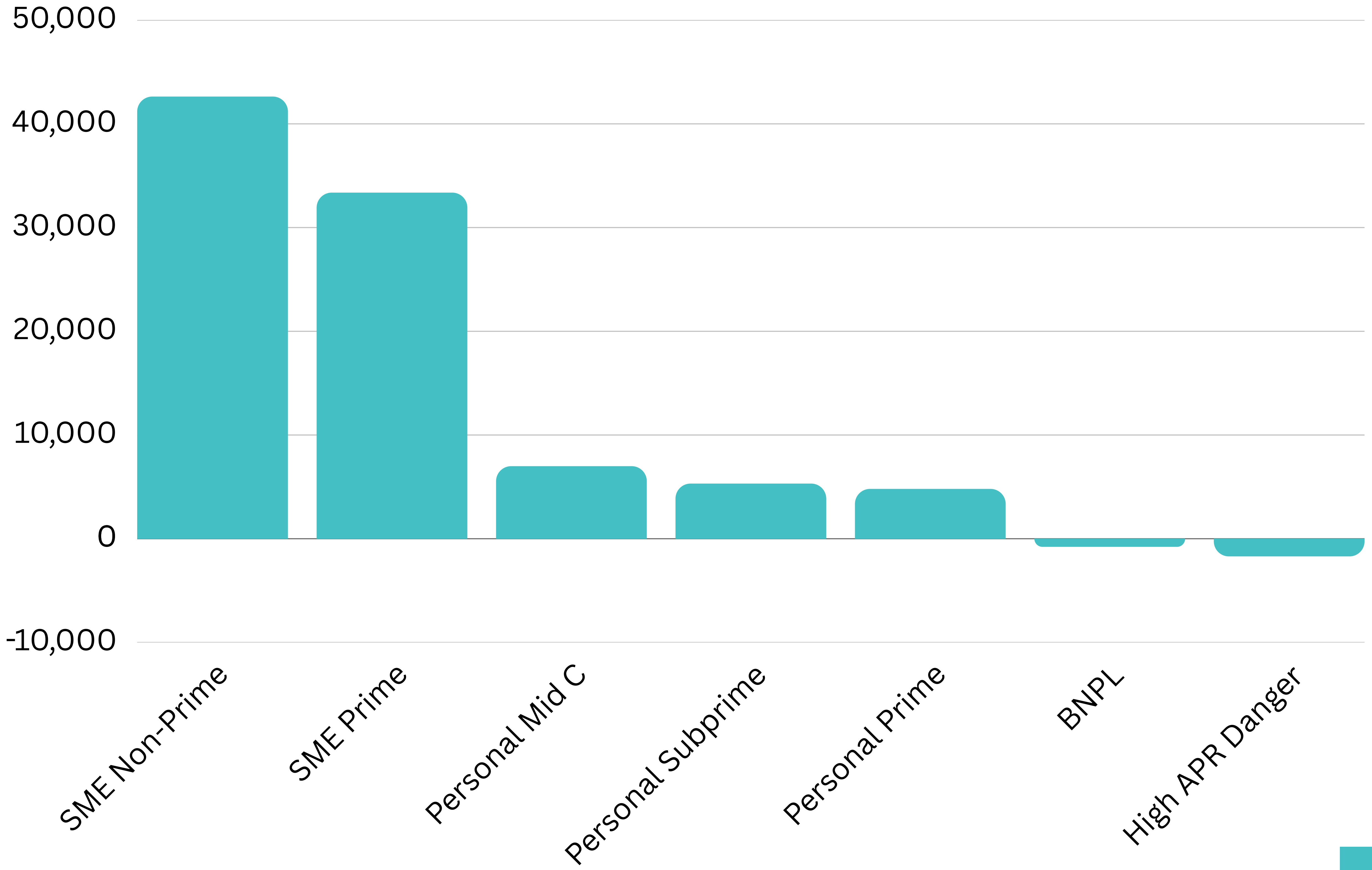
Combined impact: +₹17.85 cr at baseline → +₹18.80 cr at +3pp stress · volume −₹58.18 cr (~4.2%) · ***Recommendation strengthens under stress.***

Context & Problem

Losses don't spread evenly, they concentrate in two segments that together make over a third of the book.

- 49,600 Loans
- ₹1,398 cr origination
- 7 segments

The recommendation operates on this concentration, the loss-makers can't be repriced into health.
Cleanest segment is the one with room to grow



Source: WS E step6_hybrid_metrics.csv · 7-segment hybrid segmentation · origination total per WS I: Section 2.

Approach

Data → Segments → Early Warning → Policy.
One number, one source and every figure reconciles.

1
Build & validate

- 40,000 customers
- 49,600 loans
- 381,032 repayments
- 350,187 behaviour-month rows.
- Generated against a locked data-design spec
- **Gate 1 verdict: PASS · 32 / 32 MUST checks.**

2
Diagnose

- Channel economics across 5 acquisition channels (Referral, Partner-embedded, Digital ads, Organic, DSA).
- 7-segment hybrid segmentation that separates risk and value coherently.

3
Model & monitor

- Behavioural early-warning scorecard, validated out-of-time: **Gate 4 PASS, AUC 0.883 on currently-clean loans. .**
- Four candidate policy levers evaluated at baseline, +2pp, and +3pp default stress.

4
Cost & stress-test

- Sensitivity tested across LGD ±10pp, CoF ±1pp, cash-flow cut-off 0.5–0.7, growth 5–20%.
- Recommendation strengthens monotonically under stress.

Conventions followed by each figure:

Default = ever 90+ DPD · observation window 18-24 months · LGD 60% retail / 50% SME · CoF 11.0% · opex ₹80/active loan-month · macro stress band +2 / +3pp · ranges, never point estimates · one workbook, one source.

Portfolio Diagnostic

Inside otherwise-profitable segments, a single behavioural signal cleanly separates loss-makers from the rest

The cut-off: cashflow_consistency_mean below ~0.6 (Personal) / 0.65 (SME) — a signal already derived from bank statements pulled at application via account aggregator.

Segment	Slice Declined	Slice Kept (Untouched)	Segment Overall
SME Non-prime C-D-E	• ~950 apps/yr • 17% of flow • ₹88.67 cr origination foregone	• ~4,500 loans/yr • profitable, unchanged	• 5,461 loans · 8.05% default • +₹42,631 / loan • +₹22.37 cr total
Personal Subprime D-E	• 767 loans/yr • 13% of flow • ₹9.39 cr foregone	• ~4,700 loans/yr • profitable, unchanged	• 5,486 loans • 13.51% default • +₹5,318 / loan • +₹2.99 cr total

Personal Subprime D-E has the highest default rate in the book (13.5%) but is still profitable on the average loan so it doesn't need exiting, it needs trimming.
SME Non-prime carries large tickets (~₹9 lakh average), so the same rule has 5-6× the rupee impact.

Recommendation is to "contain" rather than "exit" by removing the identifiable loss-makers and leaving the profitable ~85% alone.

WS E step6_hybrid_metrics.csv (segment economics) · WS I §3 (SME C-D-E rule, ₹88.67 cr, ~950 apps) · WS I §4 (Personal D-E rule, ₹9.39 cr, 767 loans) · WS H cutoff_sensitivity.csv (0.6 confirmed as empirical optimum).

Default isn't uniform and it ranges 17× from grade A to E. Risk grade is the spine of policy.

GROW
low risk · high value
Expand profitable segments to absorb volume shed elsewhere.

SME Prime (A-B) — 9.2% of book
4,558 loans · 3.1% default · +₹33,369 / loan

→ **Lever Grow originations +10% · +₹1.47 cr**

CONTAIN
elevated risk · still profitable
Surgically decline the loss tail; keep the profitable majority.

SME Non-prime (C-D-E) — 11.0% of book
5,461 loans · 8.1% default · +₹42,631 / loan
→ **Headline lever · +₹12.97 cr**

Personal Subprime (D-E) — 11.1% of book
5,486 loans · 13.5% default · +₹5,318 / loan
→ **Supporting lever · +₹2.33 cr**

MAINTAIN
low risk · moderate value
No policy change; the working core of the book.

Personal Prime (A-B) — 20.1% of book
9,988 loans · 4.3% default · +₹4,803 / loan

Personal Mid (C) — 13.1% of book
6,513 loans · 7.5% default · +₹6,996 / loan

→ **No lever · 33.2% of book unchanged**

EXIT
loss-making per loan
Stop new acquisition; let existing book run off.

BNPL (APR ≤ 32.4%) — 26.6% of book
13,180 loans · 6.5% default · −₹782 / loan
→ **Cease Digital ads + DSA · +₹1.08 cr**

High-APR Danger (>32.4%) — 8.9% of book
4,414 loans · 42.1% default · −₹1,703 / loan
→ **Effectively closed · let run off**

Seven segments · 49,600 loans · ₹1,398 cr origination

Source: WS E step6_hybrid_metrics.csv · WS I Sections 3 & 6

Loss-makers are identifiable before they default, AUC 0.883 on currently-clean loans, validated out-of-time

One model, four levers. Same engine powers Contain, Exit, and the decline rules on slide 6.

DIVISION
does the model separate the bad from the clean

AUC 0.883OOT *OOT, currently-clean loans only*
KS 0.681 · top-decile lift 7.6×
Bootstrap 95% CI [0.886, 0.933]

Full-book 0.957 rejected — inflated by already-bad loans

LEAD TIME
do we have time to act before default

Median 91 days · mean 108 days
Window-capped true floor, not ceiling
Score rises monotonically into default

A quarter to restructure, decline, or contain

SEGMENT FIT
does it work where Contain actually runs

Personal Non-prime: AUC 0.896 (highest)
Distress cohort: 17.6× lift, 87% RED/Escalation
Subprime BNPL: 57% already escalated

Validates slide 6 segment by segment

OPERATIONS
can it run at portfolio scale

Watchlist: 3,890 loans/mo → catches 87%
RED 525/mo · AMBER 3,365/mo · GREEN 13,158/mo
13 behaviour-led features · monthly refresh

Source: WS F – Early Warning System · Gate 4 PASS (all 6 checks) · Seed 20260603, deterministic · OOT split: purged + embargoed

Channels & Unit Economics

No channel is broken but channel averages hide where the money actually leaks.

No channel exits required. Reallocate spend toward Organic/Referral and tighten DSA intake at grade.

Channel	Loans	CAC	Default	Profit/Loan	Payback	CLV/CAC
Referral	7,500	₹306	7.0%	₹10,086	0.55 mo	33.7x
Partner-embedded	12,359	₹613	8.0%	₹9,959	1.07 mo	17.0x
Digital ads	17,134	₹1,222	8.0%	₹9,721	2.14 mo	8.8x
Organic	5,039	₹152	7.0%	₹9,018	0.30 mo	60.0x
DSA	7,568	₹1,021	8.0%	₹8,314	1.80 mo	8.9x

DSA AS A LEVER

- Within-grade default gaps: all under 1 pp
- DSA brings 3.7 pp fewer Grade-A whereas 2.4 pp more Grade-E
- The problem is who DSA brings

WATCH ON DIGITAL ADS

- Mean ₹9,721 vs median ₹2,965 → 228% skew
- 3rd by mean profit, but LAST by median profit
- Healthy headline driven by a concentrated tail

Source: WS D Q2 Channel Economics workbook · CLV/CAC = Σ margin ÷ Σ CAC · Gate 1 PASS, 32/32 MUST checks

Product / Ticket / Tenure

One cell carries half the book and one product loses money everywhere it's sold.

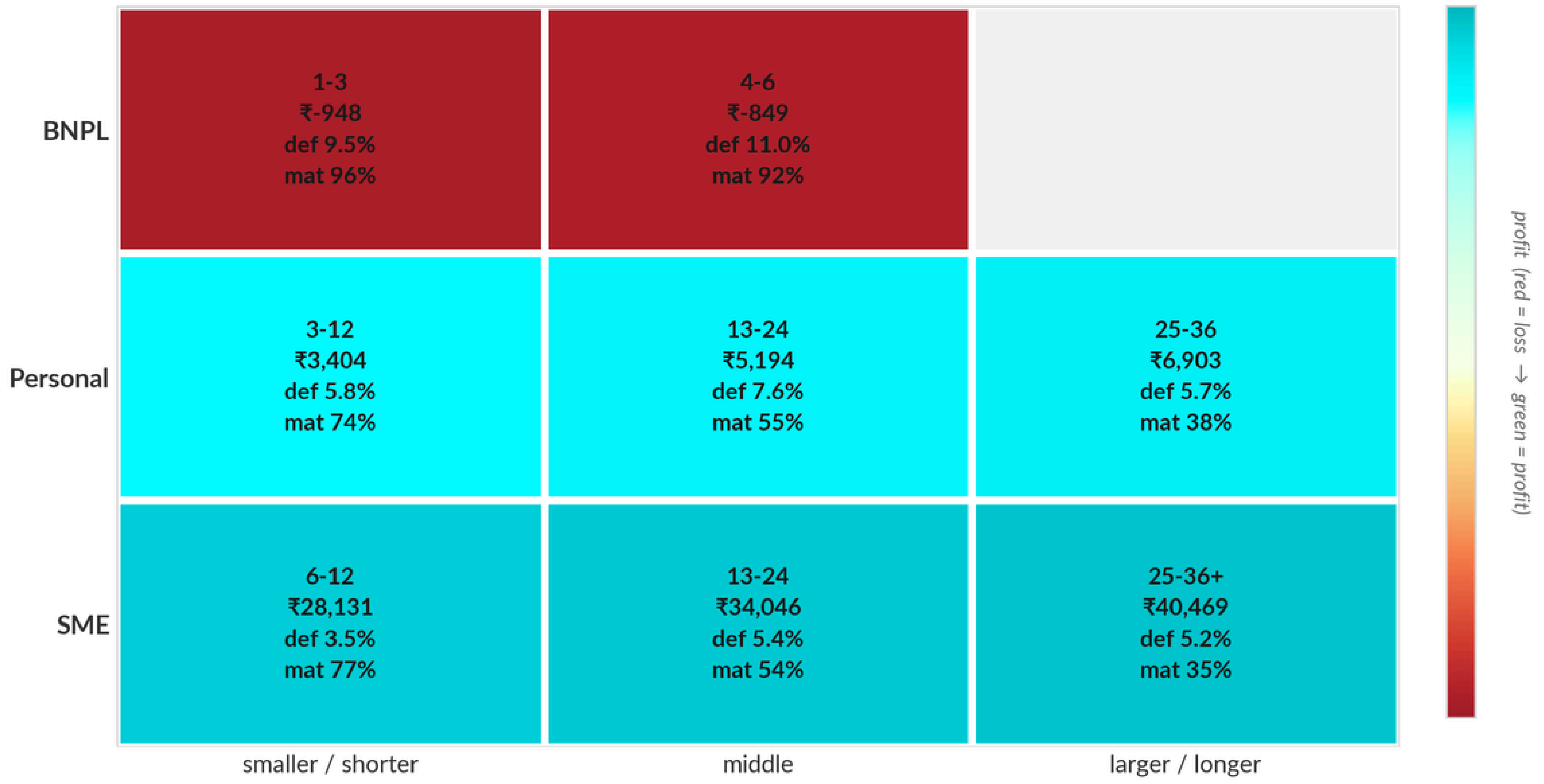
Profit rises with ticket and tenure; long tenure is genuinely lower-risk.

Slice C — Mean profit per loan by product × ticket

BNPL underwater at EVERY ticket band; large-ticket SME is the crown jewel.

Slice B — Mean profit per loan by product × tenure

mat% = how aged the cell is (low = profit is "so far", understated).



1. SME large-ticket wins

- **SME >₹1M** earns ₹69,604 paper loan
- **₹25.6 cr** over half of all book profit
- This is where growth replaces shed volume

2. BNPL = Problematic

- Loses at every ticket and tenure
- −₹803 even above ₹25k, a ticket floor won't fix it
- Total drag on the book: −₹1.57 cr

3. Cut off where it loses

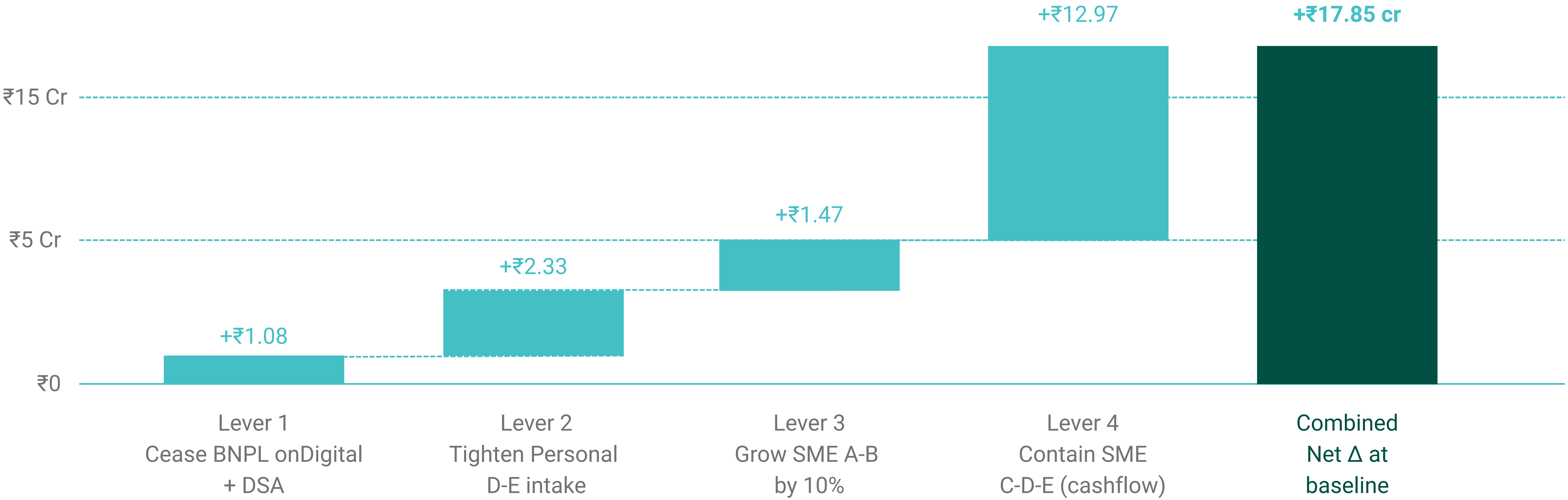
- Digital ads alone = ~50% of the BNPL drag
- Partner-embedded + DSA = ~89%
- Contain BNPL on these channels first

Source: wsD_Q2_controlled_workbook · Channel_Economics · DSA_Decomposition (footnote)

Integration of levers

All four levers combined add +₹17.85 cr which gets stronger under stress

Defensive levers gain more as defaults rise while growth lever cedes some upside but stays positive at +3pp shock.



STRESS TEST

Baseline · +₹17.85 cr
+2pp default shock · +₹18.48 cr (+0.63)
+3pp default shock · +₹18.80 cr (+0.95)

ACTS AS A COUNTER

Defensive levers save more as defaults rise L5 alone: ₹12.97 → ₹14.30 cr at +3pp Stronger downside protection, not weaker

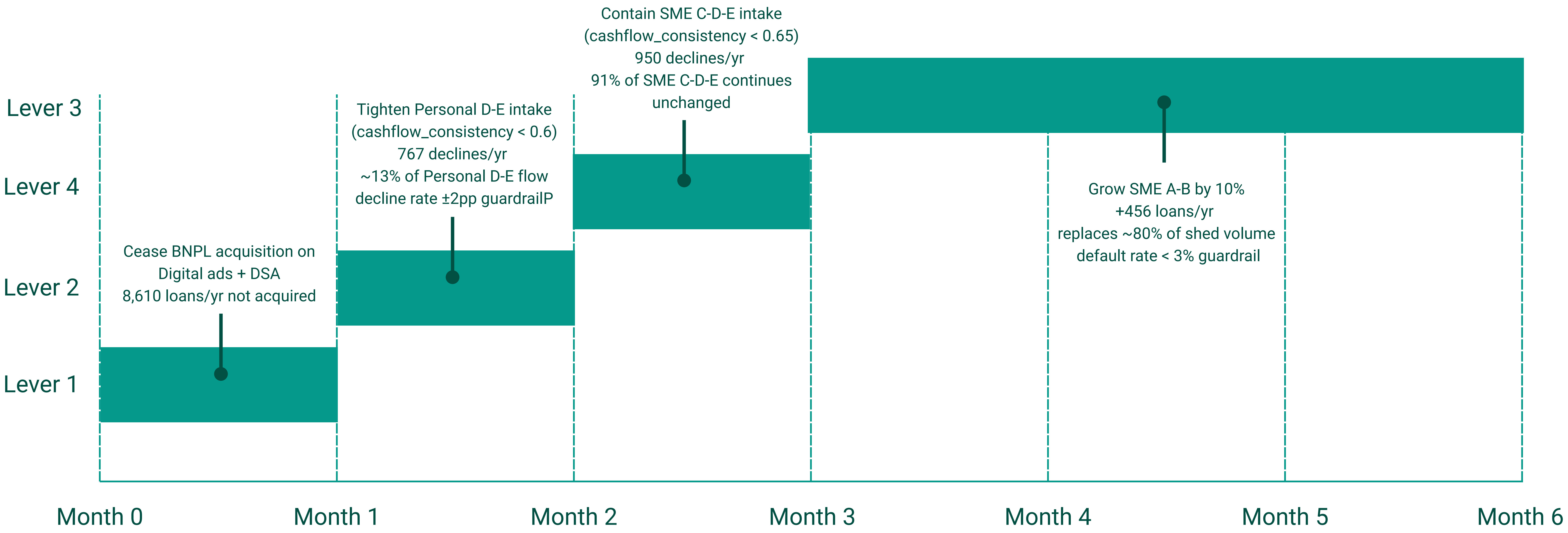
VOLUME TRADEOFF

Origination shed: −₹58.83
Contribution gained: +₹17.85 cr
~30% margin on the volume cut

Recommendations

Each lever, costed and guard-railed, the implementation view.

Five levers, named owners, named guardrails, quantified rupees.



Early Warning Engine — in-life detection (WS F) is already live
Watchlist 3,890/mo , 87% recall , mean 108-day lead time

Operational with a ₹3-5 cr provisional impact

Monitoring and Implementation Surface

Monitoring with named escalation triggers while no new infrastructure is built underneath

HOW WE'LL KNOW IT'S WORKING

Monthly - executive dashboard

- Combined contribution Δ vs straight-line plan
- Decline rates on L2 (Personal D-E) and L5 (SME C-D-E)
- L4 ramp progress against 10% growth target
- EWS recall · default-rate drift by product

Weekly - Risk Ops control surface

- Watchlist movement · RED action queue
- Signal pipeline health · scoring latency
- False-positive watch · decline-gate firing rate

WHAT THIS RECOMMENDATION DOES NOT REQUIRE

- No new data sources or aggregator changes
- No model retraining
- No infrastructure migration
- No vendor changes
- No new headcount in Risk

WHEN WE'LL ESCALATE

- • Combined Δ < 80% of plan for 2 months
- Decline rate outside ±2pp band (L2 or L5)
- L4 ramp behind by >30% pace beyond M4•
- EWS recall <80% or signal coverage <95%•
- Any product default drifts >1pp in a month